

Saanidhya Shahi

AIML B2

09619051623

Fuzzy Logic

Exp. - Fuzzy SVM

Code -

% Fuzzy SVM Implementation in MATLAB

% Corrected version for lab assignment

clc; clear; close all;

%% Step 1: Generate synthetic dataset

rng(1); % reproducibility

X = [randn(50,2)*0.75+ones(50,2); randn(50,2)*0.75-ones(50,2)];

Y = [ones(50,1); -ones(50,1)];

%% Step 2: Assign fuzzy membership values

mu = zeros(size(Y));

meanPos = mean(X(Y==1,:));

meanNeg = mean(X(Y==-1,:));

for i = 1:length(Y)

if Y(i) == 1

dist = norm(X(i,:) - meanPos)^2;

mu(i) = exp(-dist);

else

dist = norm(X(i,:) - meanNeg)^2;

mu(i) = exp(-dist);

```

end

end

%% Step 3: Train Fuzzy SVM with weights
SVMModel = fitsvm(X,Y,'KernelFunction','rbf','BoxConstraint',1,'Weights',mu);

%% Step 4: Plot decision boundary
d = 0.02;
[x1Grid,x2Grid] = meshgrid(min(X(:,1)):d:max(X(:,1)), min(X(:,2)):d:max(X(:,2)));
xGrid = [x1Grid(:),x2Grid(:)];
[~,scores] = predict(SVMModel,xGrid);

figure;
gscatter(X(:,1),X(:,2),Y,'rb','o^');
hold on;
contour(x1Grid,x2Grid,reshape(scores(:,2),size(x1Grid)),[0 0],'k','LineWidth',2);
title('Fuzzy SVM Decision Boundary');
xlabel('Feature 1'); ylabel('Feature 2');
legend('Class +1','Class -1','Decision Boundary');
grid on;

%% Step 5: Evaluate performance
CVSVMModel = crossval(SVMModel);
classLoss = kfoldLoss(CVSVMModel);
fprintf('Cross-validated classification loss: %.4f\n', classLoss);

```

Output -

